Rainfall Estimation Using Machine Learning Approaches with Raingauge, Radar, and Satellite Data

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***Abstract*— Rainfall is an essential variable in studying weather and has a vital role in the hydrological cycle. Therefore, accurate rainfall information becomes an urgent need. Machine learning approaches have also shown remarkable developments in various areas of life including rainfall estimation. This research combined rain gauge, radar and satellite data with several algorithms such as Decision Tree, Random Forest, Adaptive Boosting, and Gradient Boosting. Interestingly, it was found that the Gradient Boosting learning ensemble algorithm is best applied in rainfall estimation, with a correlation value (R2) is 0.740 and RMSE value is 1,765. On the other hand, weather radar reflectivity data significantly influences the integration of the data built into the model, suggesting that details of optimizing data are significantly required.**

***Keywords*— machine learning, rainfall, estimation, gradient boosting**

I. Introduction

Rainfall information has a crucial role in various areas of life. Accurate observation of rainfall is important to improve the quality of weather prediction [1]. Rainfall observations can be done in two ways: direct observation based on weather stations and indirect observations or observations based on remote sensing such as weather radar and satellites [2].

Observation of rainfall directly using a rain gauge. Unfortunately, this method provides limited spatial information, especially for convective rainfall or in areas with complex topography [3]. Meanwhile, remote sensing-based rainfall products can provide large-scale information with limited accuracy.

Reliable rainfall data may not be met if using only single data source [4]. On the other hand, in previous studies, machine learning nonparametric methods with direct and indirect measurement data sources can be used to estimate rainfall [5]–[7].

This study will discuss the application of machine learning to estimate rainfall using rain gauges, weather radar and satellite data sources. The algorithms used are Decision Tree (DT), Random Forest (RF), Adaptive Boosting (Adaboost), and Gradient Boosting (GB). The four algorithms will be compared to obtain the best and most appropriate algorithm for estimating rainfall.

II. Tree Based Algorithm

One of the machine learning techniques is DT. The DT concept is to turn a pile of data into a decision that represents the rules of a decision. Currently, DT has undergone several developments called ensemble trees. One of them is RF which is a development consisting of a combination of several DTs. Another tree ensemble development is the application of bagging and boosting. The development of a tree-based algorithm is shown in Figure 1.

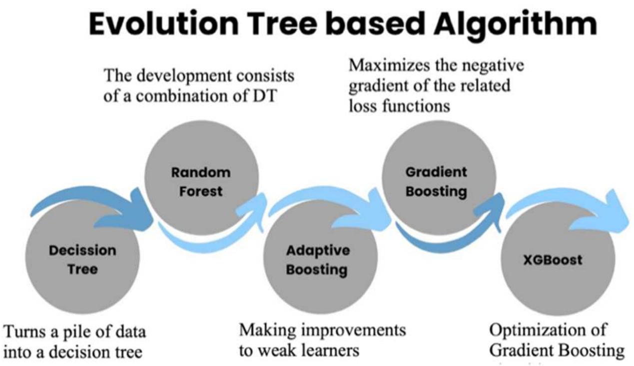

Fig. 1. Evolution of tree-based machine learning algorithms.

A. Decision Tree

The DT terminology starts with a single node or nodes, and then the node branches to represent the available options. DT is like a tree with roots, branches, and leaves, as shown in Figure 2. DT follows the same structure; a root node is the parent of all the top nodes in the Tree. Each node shows a feature (attribute), each link (branch) shows a decision (rule), and each leaf shows a result (categorical or continuous value) [8]. DT creates all possible alternatives and traces them to a conclusion in one view. That is to facilitate comparisons between the various alternatives [9]. Non-linear data does not affect any parameter of DT. One of the advantages of DT is that it is transparent and easy to interpret.

In principle, DT types can be grouped into regression and classification trees that can be used to solve regression and classification problems. It accepts categorical and numerical data input differ. A regression tree differs from linear regression, which can only accept numerical data input. The regression tree does not use assumptions as a relationship

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between dependent and independent variables like linear regression. There is no mathematical equation generated by the regression tree [10].

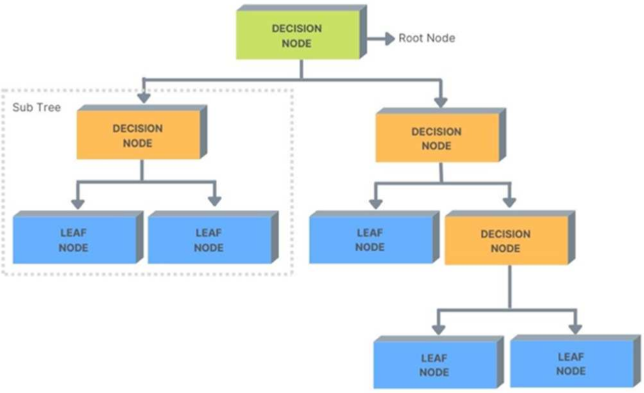

Fig. 2. DT Algorithm flowchart is like an inverted tree, where the roots are at the top while the branches and leaves are at the bottom.

In sequence, the tree formation process goes through the stages of determining the root node, placing the dataset on the root node, splitting the dataset into subsets, determining the decision node, and repeating the process until the stopping criterion is reached. DT identifies the problem by selecting the attributes tested on each node. Information Gain (IG) is a statistical value used to select attributes that expand and generate new nodes in the tree algorithm. IG contains the amount of information obtained about random variables from other random variables observed. IG is obtained from entropy, the average level of information generated by stochastic data sources. Entropy is nothing but a measure of disorder. Entropy is used as a parameter to measure the heterogeneity of the data sample set—the more heterogeneous, the greater the entropy. The entropy value always lies between 0 and 1. The value is better when it is closer to or equal to 0. Figure 3 shows a picture of entropy.

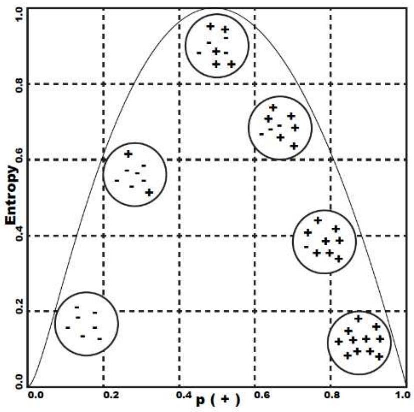

Fig. 3. Entropy. The x-axis shows the proportion of positive data points, and the y-axis shows the entropy.

Entropy is lowest when the data does not contain positive examples or only positive examples. This means that the irregularity is 0. The entropy is highest in the middle when the data is evenly distributed between the positive and negative examples. Mathematically it can be written as:

$$Entropy\ (S) = \sum_i^c P_i \log_2 P_i \qquad (1)$$

where c is the number of values in the target attribute, and P_i is the sample for class i.

IG is used as attribute effectiveness in classifying data calculated based on entropy with the following equation:

$$Gain\ (S, A) = \sum v \in V(A) \frac{|S_v|}{|S|}\ Entropy\ (S_v) \qquad (2)$$

where A is an attribute, V represents a possible value for attribute A. The meaning of $V(A)$ is the set of possible values for attribute A. $|S_v|$ is the number of samples for the value of v, and $|S|$ the number of all data samples, and $Entropy\ (S_v)$ is the entropy for samples that have a value of v.

The number of branches in a tree indicates the presence of noise in the training data. Tree pruning removes the branches so that it becomes easier to understand. It can be done with the Prunning Decision Tree approach. The Prunning Decision Tree work process in a decision tree eliminates unnecessary antecedents and rules [11].

B. Random Forest

RF is categorized as homogeneous ensemble learning because it combines several similar models. Applying a similar model with the same training set will produce the same trained model. In RF, a random sampling with a replacement method is applied to break down the data in the training set before being processed on each model, in this case, DT. Several DTs on the RF were then trained using individual samples, and each attribute was split on a tree selected between random subset attributes. RF uses a recursive binary split method to reach the end nodes in a tree structure based on classification and regression trees. Several DTs is evaluated by calculation method. Determination of the final result in the case of classification based on the majority of votes for each tree. Whereas in the case of regression, the RF results are determined from each tree's average regression value. The RF diagram is shown in Figure 4.

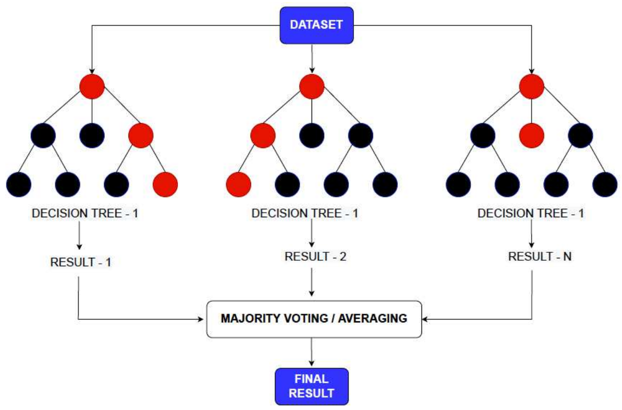

Fig. 4. Random Forest consists of several decision trees.

The technique used in the RF algorithm is a development of the bootstrap aggregating technique or often called Bagging. The difference between RF and Bagging lies in the splitting process in a tree formation. The Bagging looks for the best variable from all the existing variables, while RF only looks for the best variable from some randomly selected

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variables and then looks for the best from the selected ones. Figure 6. shows the relationship between splitting and the correlation between trees [12]. The fewer variables used in the splitting process, the lower the correlation between trees. With a more negligible correlation, it is hoped that the prediction accuracy will eventually become better.

RF is an effective method in machine learning because it can solve over-training problems that may be faced by DT [13]. The RF algorithm also has several advantages, such as low errors, good classification performance, handling large amounts of training data efficiently, and an effective method for estimating missing data and resisting outliers [14], [15].

C. Boosting

Boosting builds an ensemble model by concatenating several weak decision trees in sequence. This join assigns a weight to the output of each tree, then assigns the wrong classification of the first decision tree with the higher weight and inputs it to the next tree. After completing many itterations, the boosting method combines these weak rules into one strong predictive rule.

1) Adaptive Boosting

Adaptive Boosting (AdaBoost) is one of the earliest developed boosting models. A single decision tree is called a weak learner because of its limited capabilities. The researcher illustrates that strong learners can be obtained if several weak learners are combined [16]. The AdaBoost algorithm is a process of improving weak learners through continuous training. Figure 5. describes the calculation process and an overall picture of the AdaBoost algorithm.

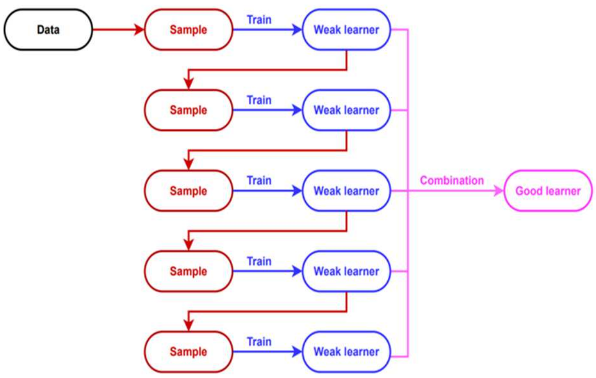

Fig. 5. Adaboost algorithm workflow.

The AdaBoost algorithm [17] begins by analyzing and normalizing the weight of the training sample, namely $d_n^1 = {}^1/_m$, m is the total sample. Iteratively then, the formation of the first tree is carried out using data with the same weight determined previously, with the number of iterations $t = 1,2 \ldots . T$. The tree in the AdaBoost algorithm is often called a stump with the character only having two final nodes obtained from one splitting, so it has low accuracy or is often called a weak learner. The resulting stump is used to predict the training data and calculate the error rate in the training dataset with the equation $\varepsilon_t = \sum_{n=1}^{N} d_n^t I(h_t(x_i) \neq y_i)$. Furthermore, ε_t will be used to calculate the weight of the votes (votes) in the final calculation of $\alpha_t = \frac{1}{2} \ln \frac{1-\varepsilon_t}{\varepsilon_t}$. The value of α_t s used to update the sample weights for the next iteration, namely $d_n^{t+1} = d_n^t \exp\left(-\alpha_t y_i h_t(x_i)\right) / Z_t$, with the normalization factor $Z_t = \sum_{i=1}^{N} d_t^i \exp(-\alpha_t y_i h_t(x))$. Finally, all processes $f(x) = \sum_{t=1}^{T} \alpha_t h_t(x)$ will be repeated until the final prediction $H(x) = sign\left(f(x)\right) = sign(\sum_{t=1}^{T} \alpha_t h_t(x))$.

2) Gradient Boosting

Gradient Boosting (GB) is a gradient-descent-based boosting algorithm derived from building with the statistical framework [18]–[20]. Similar to AdaBoost, it is a sequential training technique. The principle of the GB algorithm is to build new base learners to be maximally correlated with the negative gradient of the interrelated loss function for the entire ensemble [21], as shown in Figure 6.

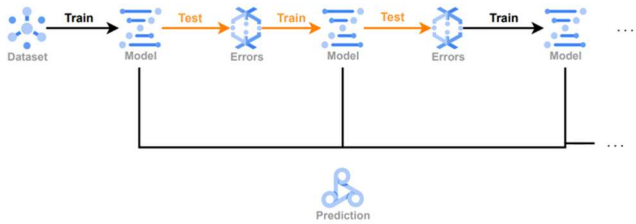

Fig. 6. Gradient Boosting algorithm workflow.

This algorithm supports high flexibility and customization with three elements: loss function, weak learner, and additive model. The GB algorithm formula is as follows:

$$F_m(X) = F_{m-1}(X) + \gamma_m h_m(X)$$
$$\gamma_m = \arg min_\gamma \sum_{i=n} L(y_i, F_{m-1}(X_i) + \gamma h_m(X_i))$$

where F_m is the model at-m from the GB method, the value of X is the input value. Meanwhile, the value of γ is the coefficient value and h_m is the decision tree at-m.

III. Dataset Dan Methodology

A. Dataset

The research location is at Radin Inten Airport, Lampung, Indonesia, with coordinates 05°14′33″S 105°10′44″E and an elevation of 86 MSL. The data used in this study are rainfall data from rain gauges, weather radar reflectivity data, and top cloud temperature data from weather satellites. The details of data are presented in Table 1. This study only uses data on rain events detected by rain gauges and weather radar. The data period used in this study is December 2021 to February 2022, as shown in Figure 7.

The data period used is focused on the rainy season in the Bandar Lampung region, namely in December, January and February. The quantity of data in the research period is also the basis for choosing the time of the study.

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TABLE I. SOURCES AND TYPES OF DATA USED IN RESEARCH

Source	Name	Time resolution	Space resolution	Unit of measurment	Description
Rain gauge	rainfall	10 min	punctual	mm	mm of precipitation
Weather radar	CMAX	10 min	400 m	dBZ	Coloumn Maximum
Weather satellite	Ch7	10 min	1 km	K	3.9µm SW IR channel
	Ch8	10 min	2 km	K	6.2µm IR(WV) channel
	Ch9	10 min	2 km	K	6.9µm IR(WV) channel
	Ch10	10 min	2 km	K	7.3µm IR(WV) channel
	Ch11	10 min	2 km	K	8.6µm LW IR channel
	Ch12	10 min	2 km	K	9.6µm LW IR channel
	Ch13	10 min	2 km	K	10.4µm LW IR channel
	Ch14	10 min	2 km	K	11.2µm LW IR channel
	Ch15	10 min	2 km	K	12.4µm LW IR channel
	Ch16	10 min	2 km	K	13.3µm LW IR channel

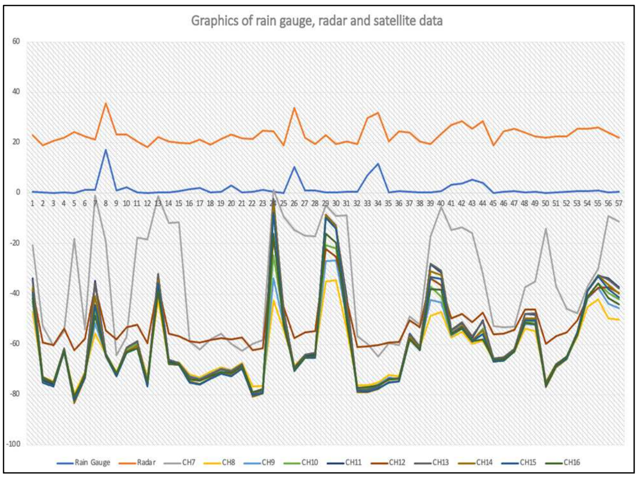

Fig. 7. Graph of the data series used in the study. The x-axis shows the number of rain events. The y-axis is the value of the data source with each unit.

B. Pre-Processing

The weather radar product used is Column Maximum (CMAX) data. In the pre-processing stage, polar coordinates are converted to Cartesian coordinates, and then interpolation is carried out to obtain data according to the observation point. The interpolation used is bilinear with the hope that no deviation is generated. CMAX data at the observation point is used for the following process. The satellite data used is Himawari-8 satellite data. It is raster data in NetCDF format. It is extracted by interpolating raster data on the selected channel to the point of observation.

C. Machine Learning Approach

This study aims to perform a comparative analysis of the Decision Tree, Random Forest, Adaptive Boosting, and Gradient Boosting algorithms used to estimate rainfall. The application used to perform the analysis is Orange Data Mining.

The process begins with selecting data using a widget. Then make a comparison of several methods that have been determined previously. Figure 8. is a widget design using a regression model on Orange Data Minning software with several selected algorithms and pre-processed input datasets.

The tuning process is carried out on each algorithm to get optimal results. The Orange Data Minning application features allow users to tune an algorithm. Tuning is done on the DT algorithm, one of which is the limit of the maximal tree depth, which is set at 100, and then the RF algorithm is the number of trees set at 10. One of the tunings in the Adaboost algorithm is the number of estimators, which is set to 15, and the regression loss function is set to linear. In the GB Algorithm, the number of trees is set to 100, and the learning rate is 0.1. The tunning feature in the Orange Data Minning application is relatively user-friendly.

The following process is to compare the regression models using test and score tools to calculate the success rate between each regression model in data mining. The training data set is 66%, with 20 repetitions of train and test.

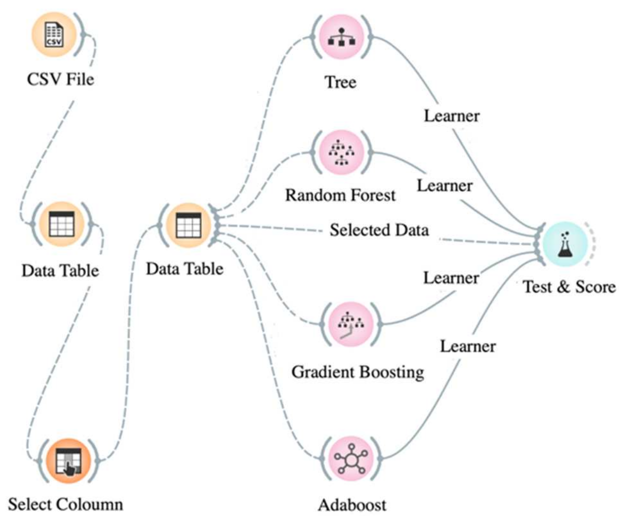

Fig. 8. The design of the application of the research model using the Orange Data Minning software.

IV. RESULT AND DISCUSSION

A. Comparison of Algorithms

By using three data sources and comparing each algorithm, it is found that the learning ensemble algorithm produces a better estimate than the basic algorithm. As shown in Table 2, the GB algorithm outperforms other algorithms from MSE, RMSE, MAE, and R2. In the orange data mining application, it is explained that:

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1) MSE measures the average of the squares of the errors or deviations (the difference between the estimator and what is estimated).
2) RMSE is the square root of the arithmetic mean of the squares of a set of numbers (a measure of the imperfection of the fit of the estimator to the data)
3) MAE measures how close forecasts or predictions are to eventual outcomes.
4) R2 is interpreted as the proportion of the variance in the dependent variable that is predictable from the independent

TABLE II. COMPARISON OF RAINFALL ESTIMATION RESULTS USING FOUR MACHINE LEARNING ALGORITHMS

Model	MSE	RMSE	MAE	R2
Decision Tree	5.148	2.269	1.271	0.571
Random Forest	3.485	1.867	1.022	0.710
Adaptive Boosting	3.256	1.804	1.005	0.729
Gradient Boosting	3.117	1.765	1.009	0.740

The results show that using the GB algorithm produces a correlation value (R2) of 0.740, greater than the other three algorithms. A value of 0.740 means a positive strong relationship exists between the rainfall variables from the rain gauge and the rainfall estimation generated using a machine learning algorithm. The RMSE value of the GB algorithm is 1,765, smaller than the other three algorithms. The GB algorithm has the highest accuracy among those three estimators due to the lowest RMSE value obtained, followed by Adaboost and RF. The ensemble learning algorithm is built from several base learners with specific techniques and has a stronger generalization ability to produce superior estimates [22], [23].

The ensemble approach is more effective in improving performance if the models tend to be independent. In Bagging, the shapes between trees tend to be similar, showing the correlation between base learners is quite significant. RF seeks to acquire different trees to increase freedom. This technique uses RF by selecting features randomly. Parts of training sets are taken as replacements for the original training sets. There are two reasons this technique is used: to increase accuracy and provide an estimate of the general error of the composite tree.

Boosting is different from bagging. The main difference is the training method. In bagging, improve the accuracy of weak learners by training multiple learners at once on multiple. AdaBoost has advantages in terms of speed, simple operation and eases of programming. No need to adjust parameters except for the number of iterations. However, AdaBoost still has drawbacks, one of which is dependent on data and weak learners [24].

The results show that the GB algorithm slightly outperforms the Adaboost algorithm. Both algorithms use the Boosting Technique with loss functions. The difference is that AdaBoost minimizes the exponential loss function, which can make this algorithm sensitive to outliers. With Gradient Boosting, any differentiable loss function can be used to be more robust against outliers than AdaBoost. On the other hand, AdaBoost minimizes the loss function related to misclassification and is best used with weak learners. This method is mainly designed for binary classification problems and can be used to improve decision tree performance. In comparison, GB is used to solve the differentiable loss function problem. So this technique is appropriate for classification and regression problems. In the case of Gradient Boosting, the drawbacks of the existing weak learners can be identified by gradients, and with AdaBoost, it can be identified by high-weight data points. Despite several differences between the two boosting methods, both the algorithms follow the same path and share similar historic roots. GB algorithm has proven to be one of the best algorithms in machine learning [25].

In the orange data mining application, there is a prediction feature to display model predictions on the data. The model prediction results on the data show that the GB algorithm outperforms other algorithms, as seen in Table 3.

TABLE III. PREDICTION PERFORMANCE SCORE

Model	MSE	RMSE	MAE	R2
Decision Tree	1.192	1.092	0.480	0.876
Random Forest	0.843	0.918	0.469	0.912
Adaptive Boosting	0.034	0.185	0.105	0.996
Gradient Boosting	0.010	0.101	0.073	0.999

A GB algorithm correlation coefficient of 0.999 represents a strong relationship. All of the points lie on the regression line with minimum errors, as shown by the RMSE value of 0.101

B. Effect of Data Integration

TABLE IV. COMPARISON OF RAINFALL ESTIMATION RESULTS WITHOUT WEATHER SATELLITE DATA

Model	MSE	RMSE	MAE	R2
Decision Tree	4.591	2.143	1.171	0.618
Random Forest	3.385	1.840	0.983	0.718
Adaptive Boosting	3.294	1.815	1.035	0.726
Gradient Boosting	3.187	1.785	1.004	0.735

In this section, we would like to investigate the effect of each data on this integration, thus understanding whether all three types of measurements are required for this rainfall estimate. By omitting one of the weather radar data or weather satellites, it can be obtained that the correlation between weather radar reflectivity and rainfall is very high. In the absence of satellite data information, rainfall estimation using weather radar reflectivity produces a correlation value that is not much different from the integration of all data, as shown in Table 4. These results indicate that the reflectivity of the weather radar can detect rain cloud convective cells and identify rain quite well.

On the other hand, in the absence of radar reflectivity data information, rainfall estimation using satellite data results in poor correlation values, as shown in Table 5. That is because there is no direct relationship between temperature measurements via satellite and measurements of rainfall falling to the earth's surface. The ability to

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estimate rainfall from satellites is highly dependent on rainfall intensity [26] , season factor [27], [28]and several other factors [29]. A mechanism is needed to find the relationship between the two measurements.

TABLE V. COMPARISON OF RAINFALL ESTIMATION RESULTS WITHOUT WEATHER RADAR REFLECTIVITY DATA

Model	MSE	RMSE	MAE	R2
Decision Tree	17.159	4.142	2.459	-0.429
Random Forest	14.812	3.849	2.022	-0.234
Adaptive Boosting	13.796	3.714	2.230	-0.149
Gradient Boosting	13.619	3.690	2.076	-0.134

V. CONCLUSION

A study of rainfall estimation using a machine learning approach with three sources of rain gauge data, weather radar, and weather satellite has been carried out. Four machine learning algorithms have been compared: DT, RF, Adaboost, and GB, which were used to estimate rainfall. The best rainfall estimates are obtained from the GB learning ensemble algorithm. Radar reflectivity has a significant influence on the results of rainfall estimation. Conversely, the ability to estimate rainfall from weather satellites highly depends on rainfall intensity, indicated by a negative relationship and high RMSE. Future research needs to optimize satellite data by first analyzing the relationship between satellite measurements and other parameters that affect rainfall values.

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